

SMART INDIA HACKATHON 2026



TITLE PAGE



VisionBridge

NATIVE ANDROID ACCESSIBILITY COMPANION

An AI-powered Android companion that helps people with low vision read, understand, navigate and stay safe — and hands over to a real volunteer the moment the AI is not sure.

PROBLEM STATEMENT ID

26196

PROBLEM STATEMENT TITLE

Student Innovation

THEME

MedTech / BioTech / HealthTech

PS CATEGORY

Software

TEAM ID

—

TEAM NAME

Prime Innovators



THE PERSON

Someone with low vision, out in an everyday situation



THE PHONE THEY OWN

VisionBridge — a native Android app, no new hardware



VOICE + CAMERA + AI

It listens, looks, understands and answers out loud



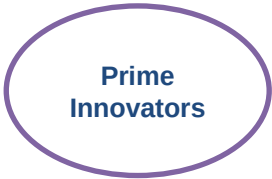
HUMAN HELP IF UNSURE

Low confidence hands over to a live volunteer



GREATER INDEPENDENCE

They read, move and decide without having to ask



IDEA / PROPOSED SOLUTION



VisionBridge — one Android app that lets a person with low vision read, understand, move and ask for help, using only their voice.

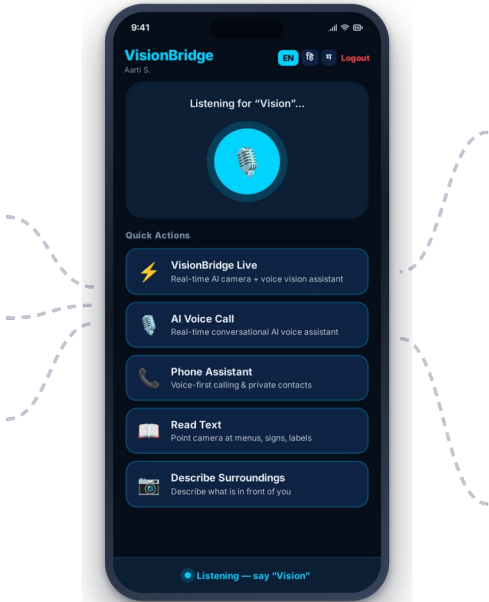
THE PROBLEM • FIVE EVERYDAY MOMENTS

-  **Printed text has no voice**
Menus, medicine labels, bills, notices
-  **Transport is guesswork**
Which bus is this? Which platform?
-  **Hazards give no warning**
A step, a parked scooter, an open drain
-  **"Where am I?"**
Street names and nearby places are invisible
-  **Always having to ask**
Independence depends on a stranger being free

 **CONFIDENCE ≥ 0.70**
The answer is spoken immediately and kept in context, so the next question needs no new photo.

THE SOLUTION • VOICE + CAMERA + AI + LOCATION + PEOPLE

-  **VOICE**
Wake word "Vision" · English, हिन्दी, मराठी
-  **CAMERA**
CameraX capture & live video frames
-  **LOCATION**
GPS · street, nearby places, SOS position



ONE ANDROID APP • 14 ASSISTANTS ON ONE HOME SCREEN

GEMINI AI
Reads, describes, decides — and scores its own confidence

HUMAN HELP
A live volunteer joins by video when the AI is not sure

 **CONFIDENCE < 0.70**
The app refuses to guess. "Shall I connect you to a volunteer?" — one spoken yes opens a live video call.

AI FIRST.
HUMAN HELP WHEN AI IS NOT ENOUGH.

HOW IT WORKS • FROM A SPOKEN SENTENCE TO A SPOKEN ANSWER

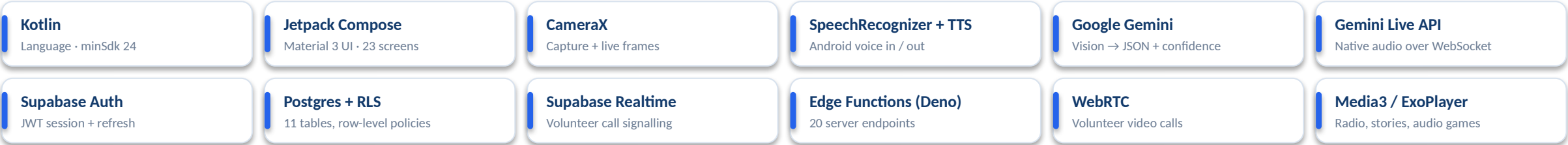


HUMAN FALLBACK Below 0.70 confidence the app stops, says so aloud and offers a live volunteer — a WebRTC video call signalled over Supabase Realtime.

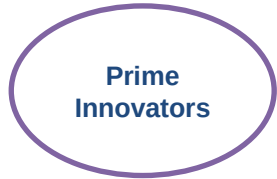
THE SAME FOUR STEPS BEHIND EVERY FEATURE



TECHNOLOGIES ACTUALLY USED IN THE CURRENT ANDROID BUILD



ALSO IN THE BUILD OkHttp · DataStore · Play Services Location · OpenStreetMap · WhatsApp Cloud API. **The Gemini key never leaves the server — Live sessions use ephemeral tokens.**



KEY FEATURES AND INNOVATION



IMPLEMENTED TODAY • 14 ASSISTANTS ON ONE HOME SCREEN

Smart Reading
Menus, labels, notices

Surroundings
What is in front of me

Currency Reader
Notes, coins, total

Object Finder
Wallet, keys, bottle

Hazard Mode
Scans every 5 seconds

Public Transport
Routes, platforms, signs

Where Am I?
Street + nearby places

VisionBridge Live
Real-time camera + voice

AI Voice Call
Conversational assistant

Phone Assistant
Voice-first calling

Volunteer Help
Live sighted helper

Emergency SOS
WhatsApp alert + GPS

Daily News
Spoken local briefing

Entertainment
Radio, stories, games

ALSO BUILT

Volunteer Dashboard
A separate login and screen for the sighted volunteer

INSIDE THE APP

Home
14 assistants, one wake word

VisionBridge Live
Camera + voice in real time

Smart Reading
Reads text aloud with a score

WHAT MAKES IT DIFFERENT • FIVE INNOVATIONS, ALL RUNNING TODAY

THE CONFIDENCE GATE

Every vision answer carries a score. Below 0.70 the app refuses to guess and offers a live volunteer.

ONE WORD, NO MENUS

“Vision” plus 41 on-device rules route most commands instantly, with no network round-trip.

LIVE NATIVE AUDIO

Gemini Live over WebSocket — camera and voice together, interruptible, on ephemeral tokens.

TRILINGUAL BY DESIGN

English, हिन्दी and मराठी across the UI, the wake word, speech and every AI reply.

RESILIENT AI LAYER

A four-model Gemini fallback chain keeps a session alive when one model is rate-limited.



WHO IT IS FOR People with low vision who already carry an Android phone — and the sighted volunteers who choose to answer when the AI is not sure.

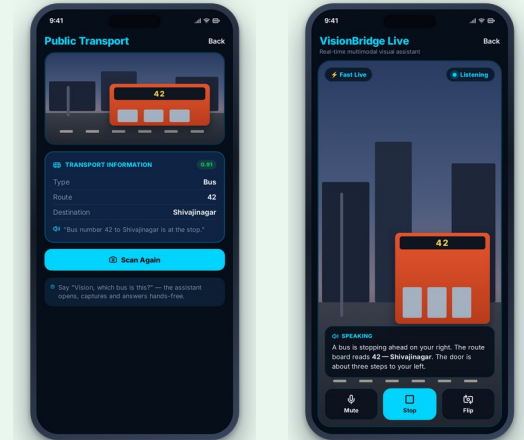
ONE REAL MOMENT • 8 P.M. AT A BUS STOP

BEFORE

- A bus pulls in. She cannot read the number.
- She looks for someone to ask — tonight nobody is there.
- The bus leaves. She waits, and asks the next stranger.
- A dedicated reading device costs money and is one more thing to carry.

WITH VISIONBRIDGE

- She says “Vision, which bus is this?” — no menus, no screen to find.
- The Public Transport assistant opens and the camera captures the board.
- Gemini reads it and the answer is spoken in her language: route 42.
- If the frame is too dark to be sure, the app says so and offers a volunteer.



WHAT CHANGES FOR THE USER



Greater independence

Reading a label or finding a bus stops depending on somebody else being free.



Fewer things to ask

Reading, transport, money, objects, location and help sit behind one sentence.



Safer daily movement

Background hazard scanning interrupts to warn before she reaches an obstacle.



Access to information

News, stories and radio arrive as speech, not as text she cannot read.



Affordable by design

No device to buy or charge — it installs on the Android phone she owns.

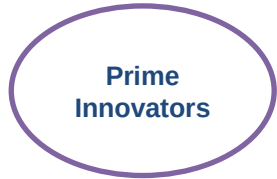


A human safety net

A live volunteer takes over instead of a confident wrong answer, in the same app.

WHAT WE DO NOT CLAIM No clinical accuracy, no guaranteed hazard detection, no offline operation, no user numbers — and no replacement for a cane, a guide or a doctor.

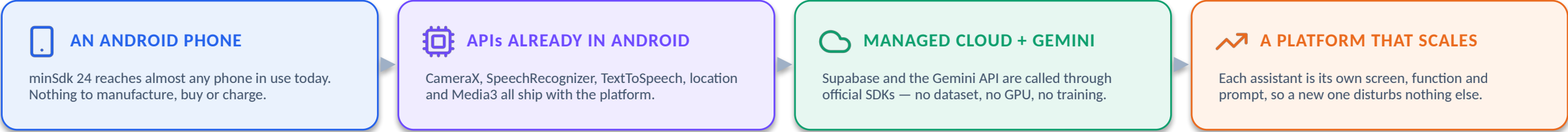
One phone. One word. A more independent everyday life.



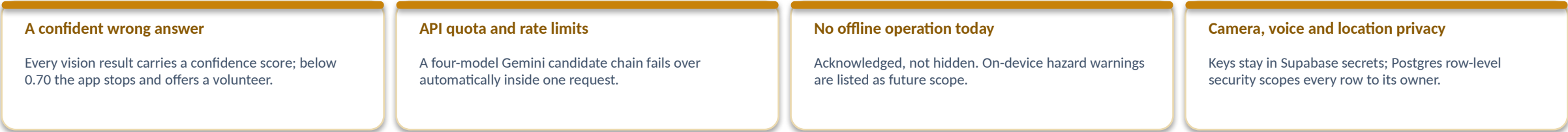
FEASIBILITY, FUTURE SCOPE & REFERENCES



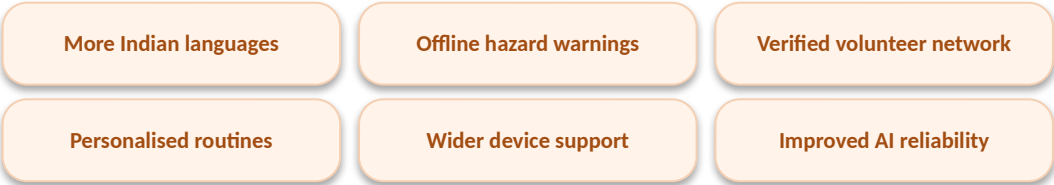
WHY IT IS FEASIBLE • NOTHING EXOTIC TO PROCURE, NOTHING TO TRAIN



RISKS WE ACTUALLY HAVE → AND HOW THE BUILD HANDLES THEM

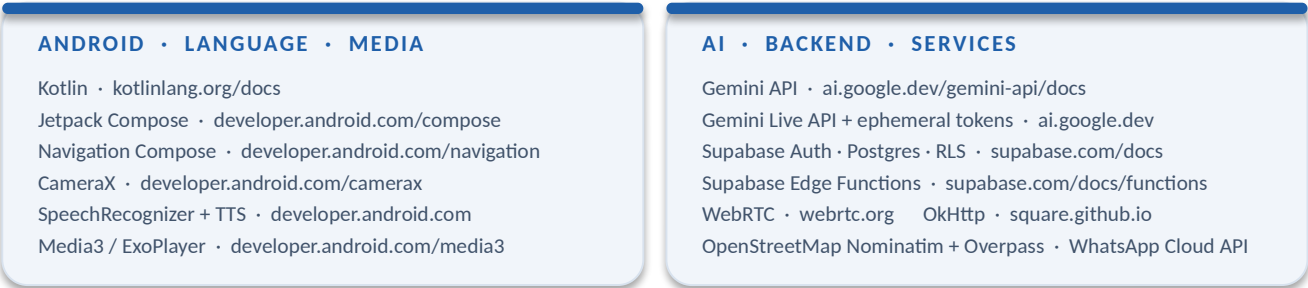


FUTURE SCOPE • NOT YET BUILT



Clearly separated from everything above, which is implemented and running.

REFERENCES • OFFICIAL DOCUMENTATION FOR WHAT IS BUILT



GitHub Repository
github.com/shailesh-pande01/visionbridge-ai

Live Demo
visionbridge-ai.vercel.app

Demo Video
[YouTube Demo Video](#)

Every reference above is the official documentation of a component already implemented in the VisionBridge Android app. No external studies, datasets or statistics are cited or claimed.